MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics
Class: Class X (CBSE)

CBSE CLASS 10 – WEEKLY TEST 06
Chapter 13: Statistics

Time: 45 Minutes
MM: 20

Document Type: Official Step-by-Step Marking Scheme **Published for:** mathsmanthan.co.in **Test Paper:** Weekly Test 06

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper units.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice & Assertion-Reason MATHS MANTHAN | (5 × 1 = 5 Marks)

Q1. If the median of a distribution is 20 and each observation is increased by 2, then the new median will be: [1]

(A) 20 (B) 22 (C) 40 (D) Cannot be determined

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) 22

Justification: If each observation in a data set is increased by a constant c, the median also increases by the same constant:

$$\text{New Median} = \text{Old Median} + 2 = 20 + 2 = \mathbf{22}.$$

$$\boxed{\text{New Median} = \mathbf{22}}$$

Q2. The relationship between mean, median, and mode for a moderately skewed distribution is: [1]

(A) Mode = 2 Median – 3 Mean (B) Mode = 3 Median – 2 Mean
(B) Median = 3 Mode – 2 Mean (D) Mean = 3 Median – 2 Mode

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) Mode = 3 Median – 2 Mean

Justification: Karl Pearson's empirical relationship connecting the three measures of central tendency states that Mode = **3** Median – **2** Mean.

Q3. If the mode of a data is 45 and the mean is 27, then its median is: [1]

(A) 30 (B) 33 (C) 36 (D) 32

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (B) 33

Justification:

$$\text{Mode} = 3\text{Median} - 2\text{Mean} \implies 45 = 3\text{Median} - 2(27)$$

$$45 = 3\text{Median} - 54 \implies 3\text{Median} = 45 + 54 = 99 \implies \boxed{\text{Median} = \mathbf{33}}.$$

Q4. For the following data, the lower limit of the modal class is: [1]

Age (years)	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50
Patients	6	11	21	23	14

(A) 20 (B) 30 (C) 23 (D) 40

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (B) 30

Justification: The maximum frequency is 23, which lies in the class interval **30 – 40**. Thus, the modal class is 30 – 40, and its lower limit is **30**.

Q5. **Assertion (A):** If the mean and median of a distribution are 15 and 16 respectively, then its mode is 18.

Reason (R): Empirical formula connecting central tendencies is Mode = 3 Median – 2 Mean. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(C) (A) is true but (R) is false.

(D) (A) is false but (R) is true.

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (A)

Justification: Using the empirical formula:

$$\text{Mode} = 3\text{Median} - 2\text{Mean} = 3(16) - 2(15) = 48 - 30 = 18.$$

Both Assertion (A) and Reason (R) are true, and (R) provides the direct mathematical justification for (A).

SECTION B: Very Short Answer Type

MATHS MANTHAN | $(1 \times 2 = 2$ Marks)

Q6. Find the median class and modal class for the following distribution: [2]

Class Interval	$0-10$	$10-20$	$20-30$	$30-40$	$40-50$
Frequency	3	8	15	7	5

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Median Class Determination [1 Mark]

Total frequency $N = 3+8+15+7+5 = 38 \implies \dfrac{N}{2} = 19.$

Cumulative frequencies: $0-10 : 3$; $10-20 : 11$; $20-30 : 26$; $30-40 : 33$; $40-50 : 38$.

The cumulative frequency just greater than 19 is 26, which corresponds to the class **$20-30$**.

$\therefore$ Median Class = **$20-30$**.

Step 2: Modal Class Determination [1 Mark]

The highest frequency is 15, which corresponds to the class interval **$20-30$**.

$\therefore$ Modal Class = **$20-30$**.

Median Class = **$20-30$**, Modal Class = **$20-30$**

SECTION C: Short Answer Type

Q7. The mode of the following distribution is 36. Find the missing frequency x: [3]

Class	$0-10$	$10-20$	$20-30$	$30-40$	$40-50$	$50-60$
Frequency	5	8	x	16	12	6

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Modal Class & Parameter Values [1 Mark]

Given Mode = 36, which lies in the class interval **$30-40$**.

$\therefore$ Modal Class = $30-40$.

Lower limit $(l) = 30$, class width $(h) = 10$.

$f_1 = 16$ (modal frequency), $f_0 = x$ (preceding frequency), $f_2 = 12$ (succeeding frequency).

Step 2: Mode Formula & Equation Setup [1 Mark]

$$\text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) \times h$$

$$36 = 30 + \left(\frac{16 - x}{2(16) - x - 12}\right) \times 10 \implies 6 = \frac{10(16-x)}{20-x}$$

Step 3: Solving for x & Final Boxed Answer [1 Mark]

Dividing both sides by 2:

$$3 = \frac{5(16-x)}{20-x} \implies 3(20-x) = 5(16-x)$$

$$60 - 3x = 80 - 5x \implies 2x = 20 \implies x = 10.$$

Missing Frequency x = **10**

Q8. Find the mean of the following distribution by the Step-Deviation Method: **[3]**

Class Interval	$50-52$	$53-55$	$56-58$	$59-61$	$62-64$
Frequency	15	110	135	115	25

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Step 1: Step-Deviation Table [1.5 Marks]

Let Assumed Mean $a = 57$, common difference between class marks $h = 3$, $u_i = \dfrac{x_i - 57}{3}$.

Class Interval	Frequency (f_i)	Class Mark (x_i)	$u_i = \frac{x_i-57}{3}$	f_iu_i
$50-52$	15	51	-2	-30
$53-55$	110	54	-1	-110
$56-58$	135	57	0	0
$59-61$	115	60	$+1$	$+115$
$62-64$	25	63	$+2$	$+50$
Total	$\sum f_i = 400$			$\sum f_iu_i = 25$

Step 2: Formula Declaration & Substitution [1 Mark]

$$\bar{x} = a + \left(\frac{\sum f_iu_i}{\sum f_i}\right) \times h = 57 + \left(\frac{25}{400}\right) \times 3 = 57 + \frac{75}{400} = 57 + \frac{3}{16} = 57 + 0.1875$$

Step 3: Final Boxed Result [0.5 Mark]

$$\boxed{\text{Mean } (\bar{x}) = \mathbf{57.19} \quad (\text{or } \mathbf{57.1875})}$$

Q9. Find the median of the following frequency distribution of examination marks: **[3]**

Marks	$15-25$	$25-35$	$35-45$	$45-55$	$55-65$	$65-75$
Frequency	8	10	15	25	40	20

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Step 1: Cumulative Frequency Table [1 Mark]

Marks	Frequency (f_i)	Cumulative Frequency (cf)
$15-25$	8	8
$25-35$	10	$8+10=18$
$35-45$	15	$18+15=33$
$45-55$	25	$33+25=58$
$55-65$	40	$58+40=98$
$65-75$	20	$98+20=118$
Total	$N = \sum f_i = 118$	

Step 2: Median Class & Parameters [1 Mark]

$N = 118 \implies \dfrac{N}{2} = 59.$

Cumulative frequency just greater than 59 is 98, which corresponds to the class **$55-65$**.

$\therefore$ Median Class $= 55-65$.

Lower limit $(l) = 55$, cf of preceding class $= 58$, frequency $(f) = 40$, class width $(h) = 10$.

Step 3: Calculation & Final Boxed Answer [1 Mark]

$$\text{Median} = l + \left(\frac{\frac{N}{2} - cf}{f}\right) \times h = 55 + \left(\frac{59-58}{40}\right) \times 10 = 55 + \frac{10}{40} = 55 + 0.25 = 55.25$$

$$\boxed{\text{Median Marks} = \mathbf{55.25}}$$

SECTION D: Case Study Based Question

MATHS MANTHAN | $(1 \times 4 = 4$ Marks$)$

Q10. CASE STUDY: COMMUNITY CLUB TAMBOLA GAME NUMBER DRAW FREQUENCY [4 Marks]

At an annual community festival, a game of Tambola (Housie) with numbered tokens from 1 to 90 was organized. The caller recorded the frequency distribution of numbers drawn across 80 completed rounds.

Number Range	$1-15$	$16-30$	$31-45$	$46-60$	$61-75$	$76-90$
Draw Frequency	8	14	22	18	12	6

(i) State the class interval showing the maximum draw frequency. [1]

(ii) Convert the intervals into continuous true limits and compute the mode. [2]

(iii) Find the cumulative frequency of tokens drawn with value 60 or less. [1]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 4 Marks]

Subpart (i): Maximum Draw Frequency Interval [1 Mark]

The highest draw frequency is 22, corresponding to the interval **31 – 45**.

$$\boxed{\text{Class Interval with Maximum Frequency} = \mathbf{31-45}}$$

Subpart (ii): Continuous Class Conversion & Mode Calculation [2 Marks]

Adjustment factor $= \dfrac{16-15}{2} = 0.5$.

Continuous intervals: $0.5-15.5$ (8); $15.5-30.5$ (14); **$30.5-45.5$** (22); $45.5-60.5$ (18); $60.5-75.5$ (12); $75.5-90.5$ (6).

Continuous Modal Class $= 30.5-45.5$.

$l = 30.5,\ h = 15,\ f_1 = 22,\ f_0 = 14,\ f_2 = 18$.

$$\text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) \times h \text{[1 Mark]}$$

$$\text{Mode} = 30.5 + \left(\frac{22-14}{2(22)-14-18}\right) \times 15 = 30.5 + \left(\frac{8}{44-32}\right) \times 15 = 30.5 + \left(\frac{8}{12}\right) \times 15$$

$$\text{Mode} = 30.5 + 10 = \mathbf{40.5} \text{[1 Mark]}$$

$$\boxed{\text{Mode} = \mathbf{40.5}}$$

Subpart (iii): Cumulative Frequency of Value ≤ 60 [1 Mark]

Sum of frequencies for tokens with value ≤ 60 ($[1-15] + [16-30] + [31-45] + [46-60]$):

$$cf(\leq 60) = 8 + 14 + 22 + 18 = \mathbf{62}.$$

$$\boxed{\text{Cumulative Frequency } (\leq 60) = \mathbf{62}}$$

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★